Solids

1 Name the following solids:

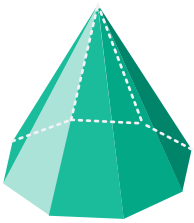
a



b



c



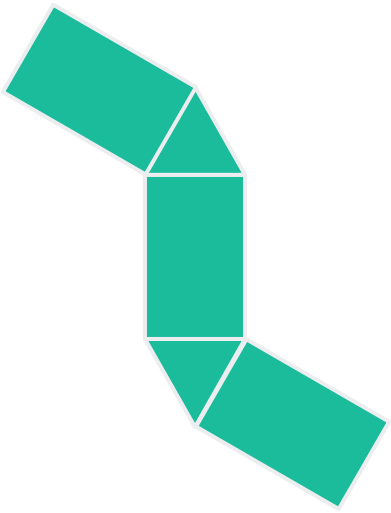
d



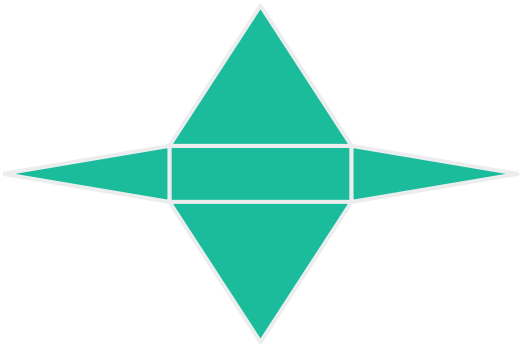
Nets

2 Name the solid formed from each of the following nets:

a



b

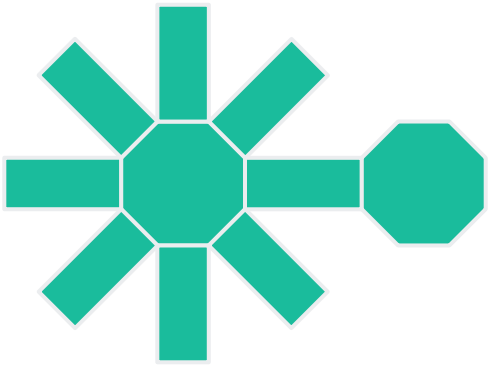


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c



d



3 Draw a net for the following solids:

a



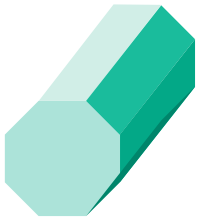
b



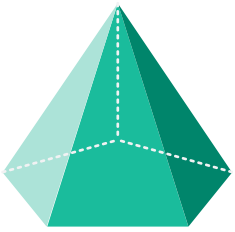
c



d



e



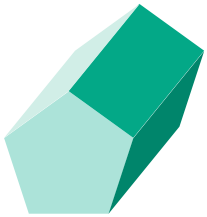
f



g



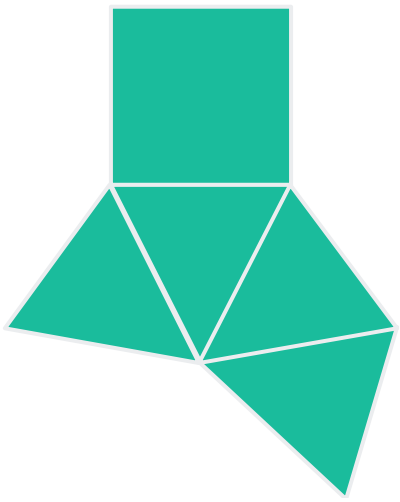
h



4 Name the 3D shapes formed by the following nets:

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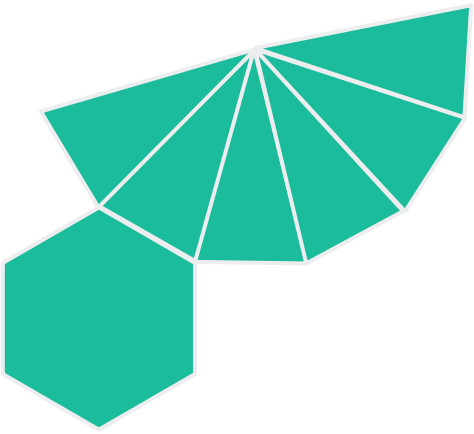
a



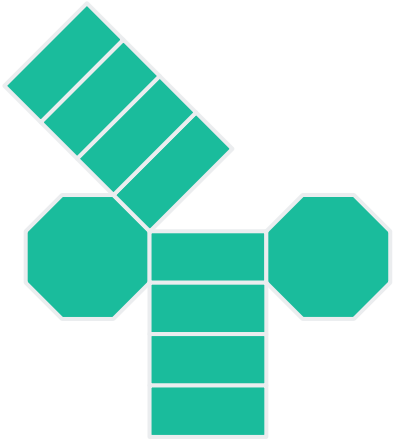
b



c



d

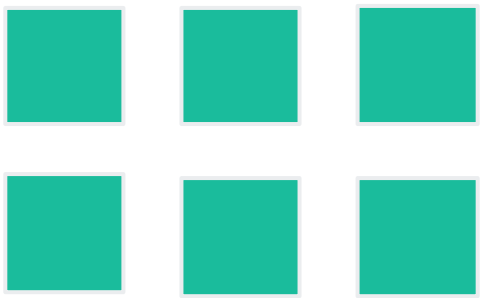


5 State the solid that can be formed from the given sets of 2D shapes:

a

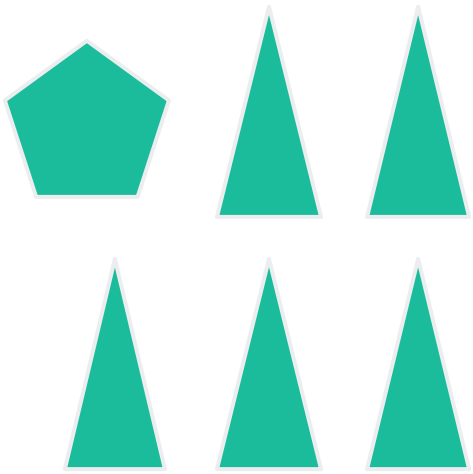


b



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c



Front, side, and plan view

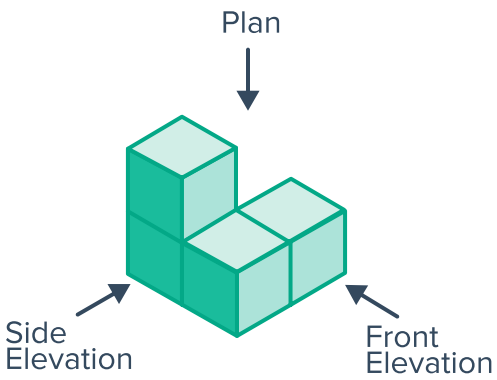
6 For each of the following solids formed from cubes:

i Draw the diagram for the side view.

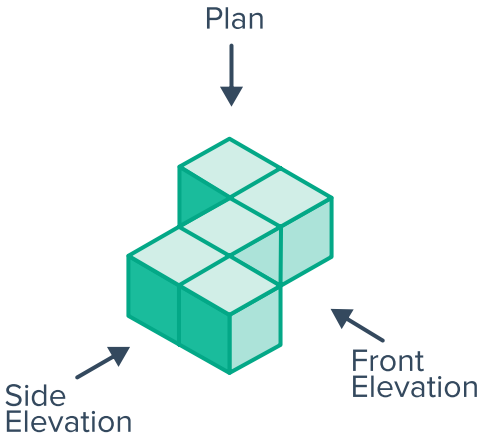
ii Draw the diagram for the front view.

iii Draw the diagram for the plan view.

a

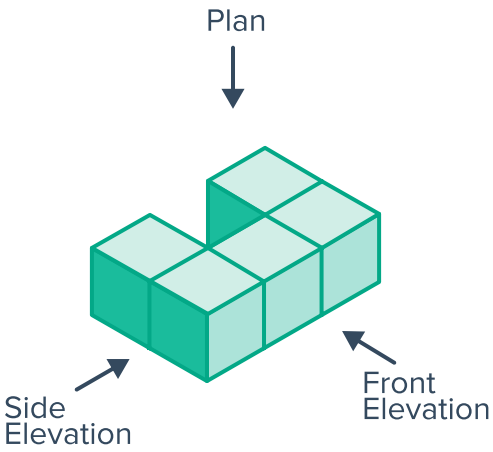


b

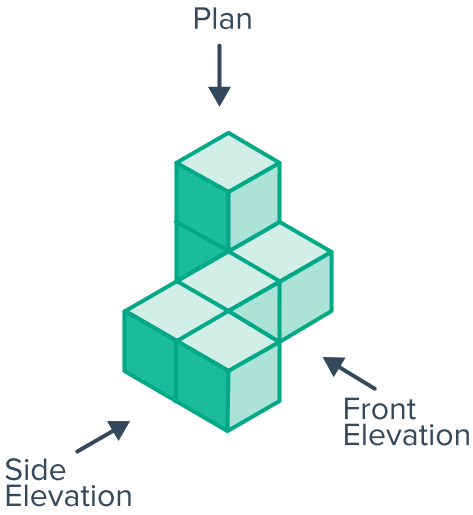


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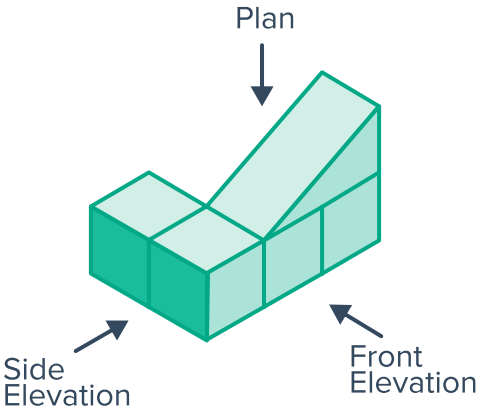
c



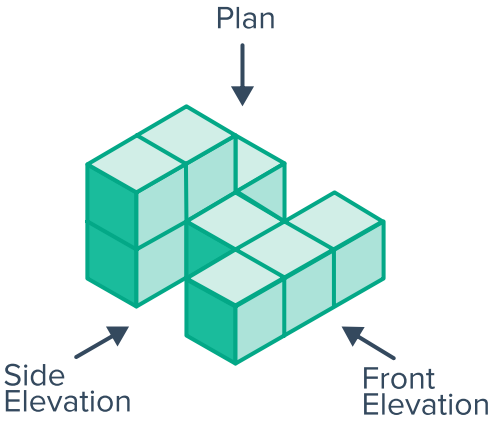
d



e



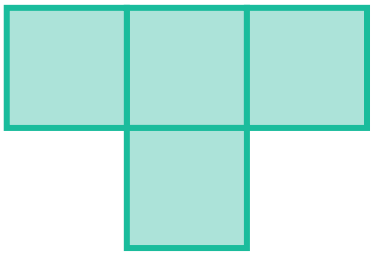
f



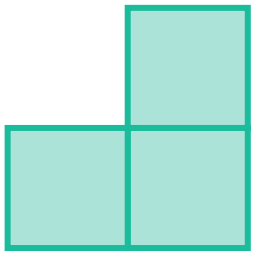
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7 Select the solid that matches the given elevations:

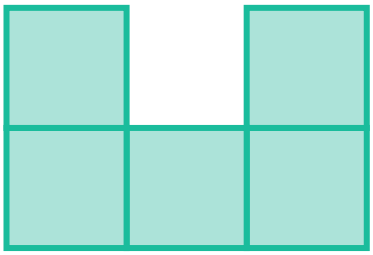
a



Plan

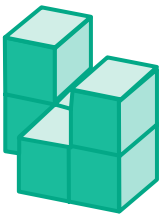


Front Elevation

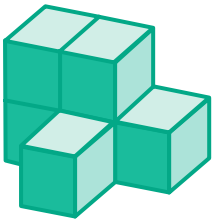


Side Elevation

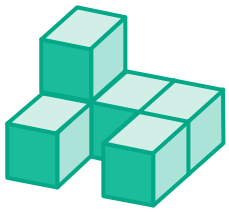
A



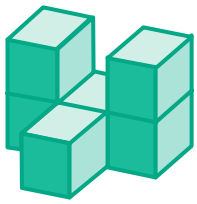
B



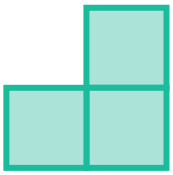
C



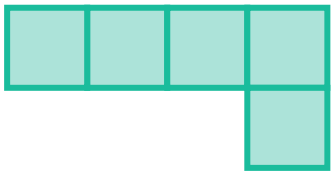
D



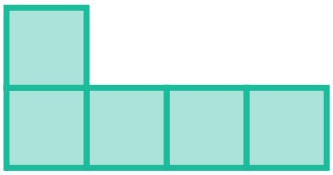
b



Front Elevation

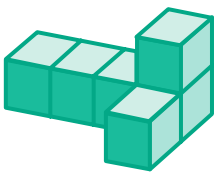


Plan

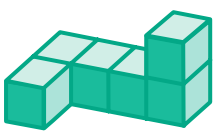


Side Elevation

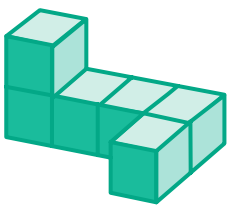
A



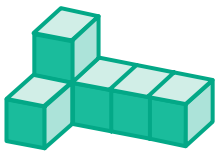
B



C

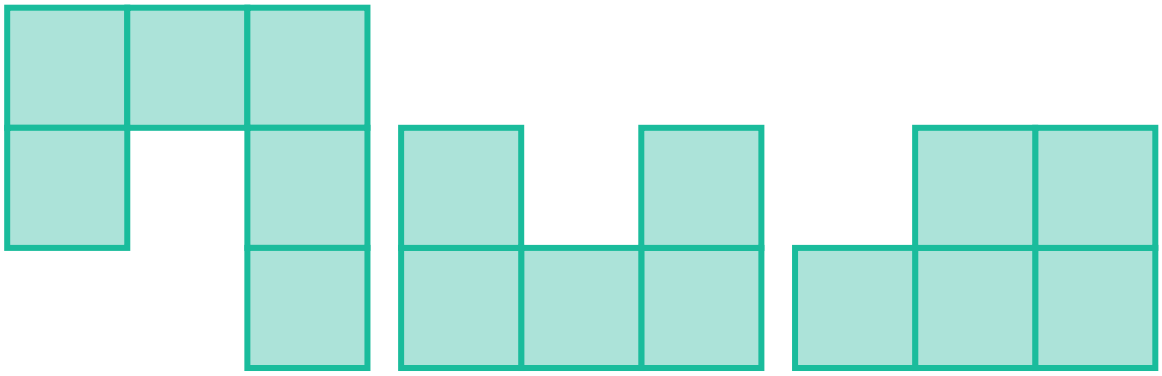


D



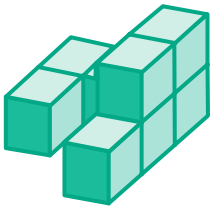
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c

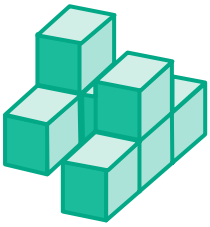


Plan Side Elevation Front Elevation

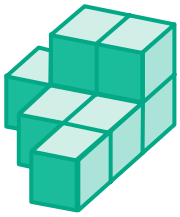
A



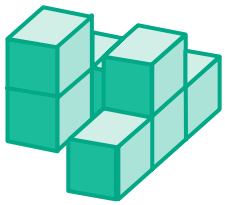
B



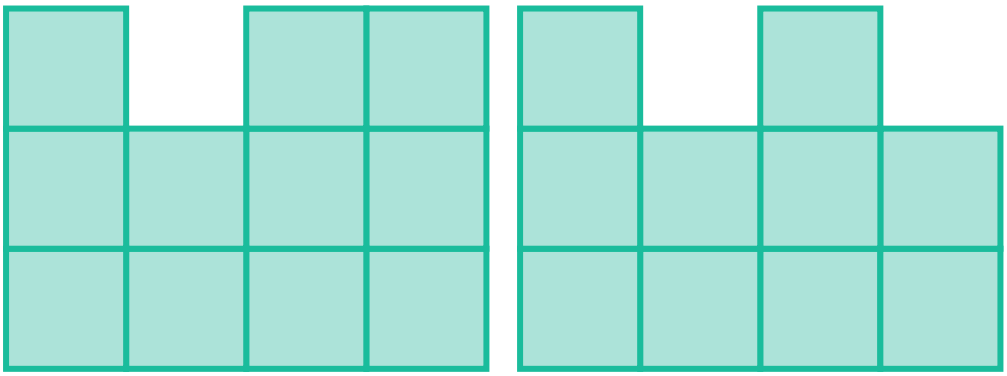
C



D

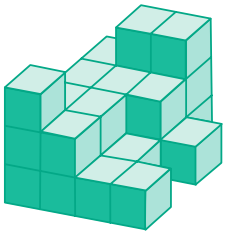


d

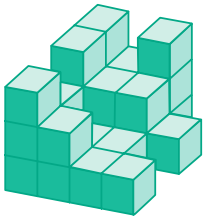


Front Elevation Side Elevation

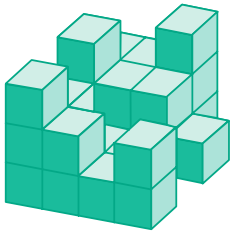
A



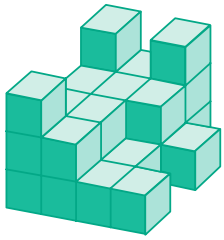
B



C



D



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8 Name the following solids:

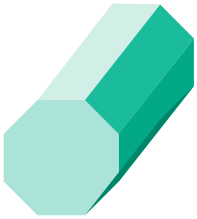
a



b



c



9 For each of the following solids:

i Name the solid.

ii State whether the solid has a uniform cross-section.

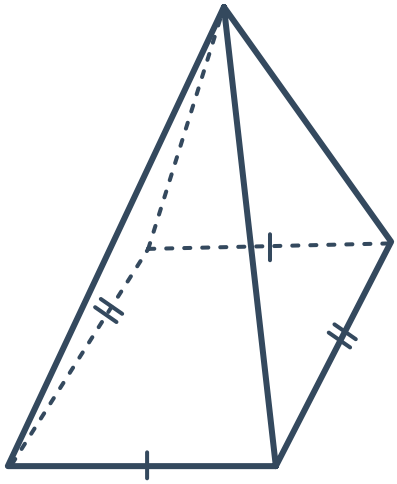
iii State the order of rotational symmetry about an vertical axis down the center of the solid.

iii State whether the solid has plane symmetry.

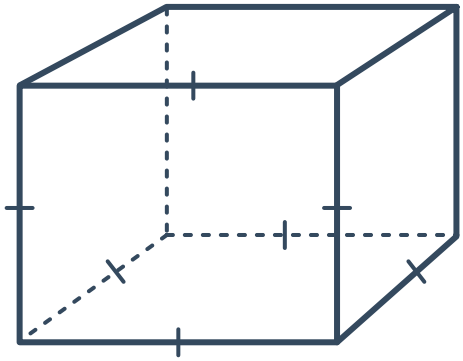
a



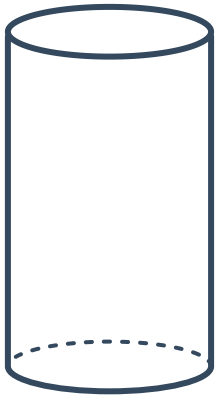
b



c

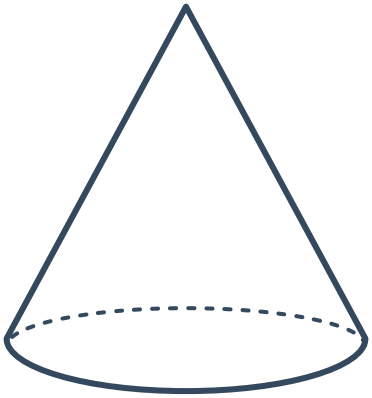


d

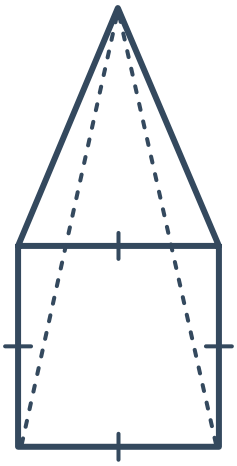


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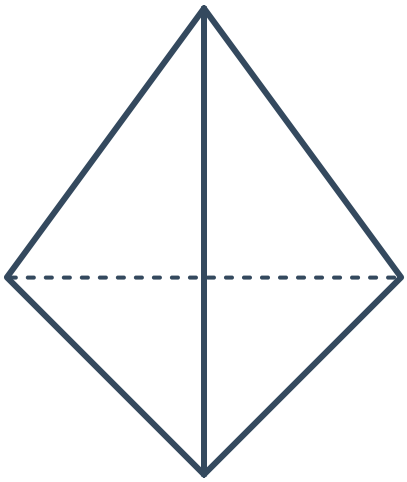
e



f



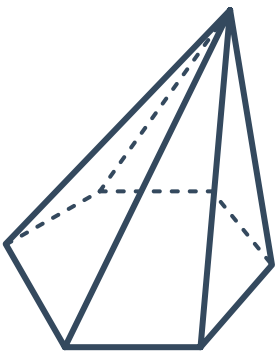
g



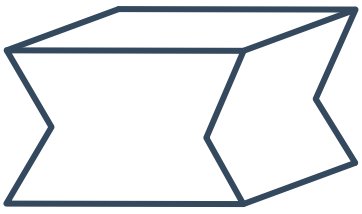
h



10 Explain why shapes A and B are polyhedra while shape C is not.



A



B

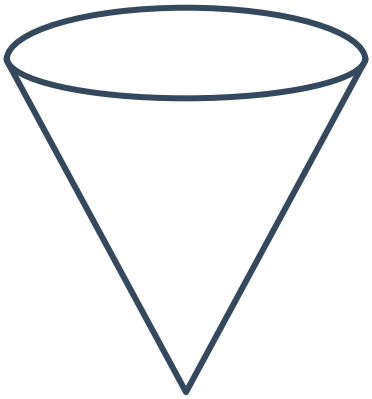


C

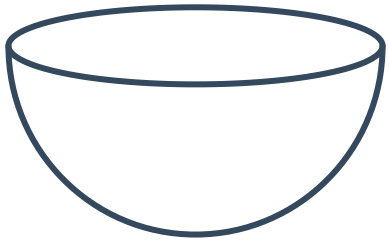
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11 Determine whether the following solids are polyhedrons:

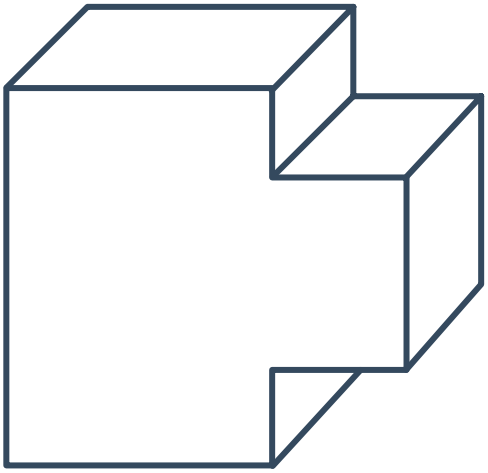
a



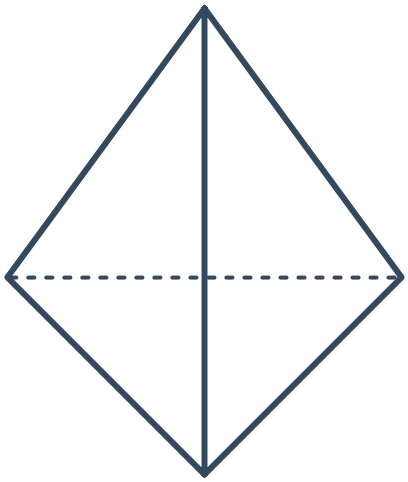
b



c



d



12 State whether the following solids have faces, vertices **and** edges:

a Sphere **b** Cone **c** Tetrahedron **d** Cylinder

13 Consider a rectangular prism.

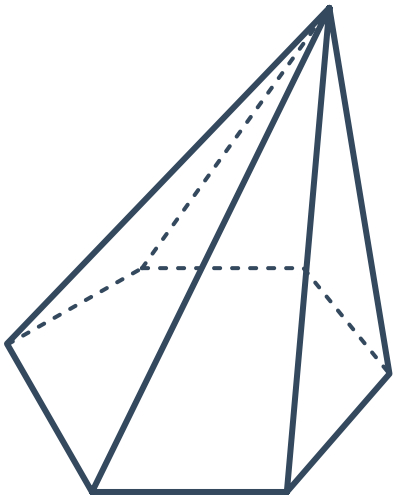
a How many faces meet at any one vertex?

b How many faces meet at any one edge?

c How many edges meet at any one vertex?

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14 Consider the following hexagonal pyramid:

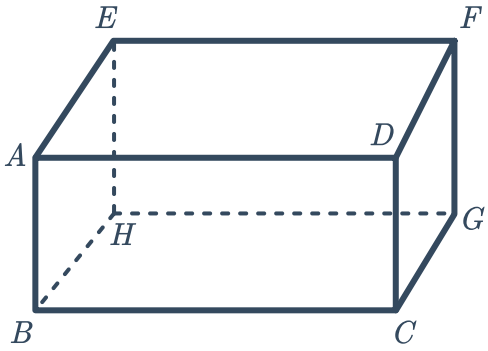


- **a** How many faces does it have?
- **b** How many vertices does it have?
- **c** How many edges does it have?

15 Complete the table with the number of features for each solid:

3D shape	Number of faces	Number of vertices	Number of edges
Hexagonal prism			
Triangular pyramid			
Rectangular prism			

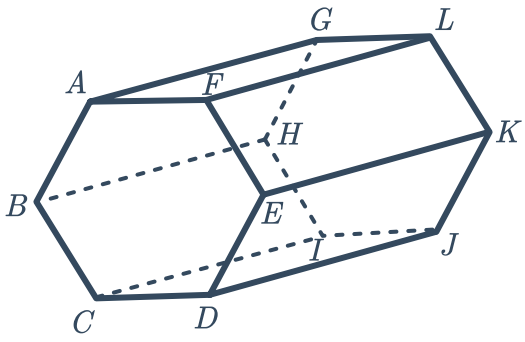
16 Consider given the rectangular prism:



- **a** Name two parallel edges.
- **b** Name two edges that are both intersecting and perpendicular.
- **c** Name all of the edges that are parallel to BH.
- **d** Name all of the edges that are perpendicular to BH.
- **e** Does the prism have plane symmetry about the plane $AFGB$?

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17 Consider the given hexagonal prism:



a Name the face that is parallel to $FEKL$.

b How many faces are perpendicular to $ABCDEF$?

c Name two parallel edges.

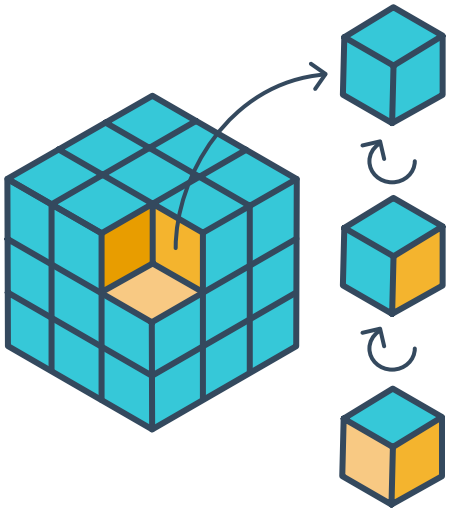
d Name two perpendicular edges.

e State whether the following are planes of symmetry for the prism:

i $BEKH$ ii $ADJG$

iii $BDJH$ iv $CFLI$

18 A number of $1\text{ cm} \times 1\text{ cm} \times 1\text{ cm}$ cubes are glued together to form a $3\text{ cm} \times 3\text{ cm} \times 3\text{ cm}$ cube as shown:



The outside faces of the large cube are painted blue and the small cubes are then pulled apart.

a How many of the small cubes will have at least one face painted blue?

b How many of the small cubes will have at least two faces painted blue?

c How many of the small cubes will have three faces painted blue?

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19 Euler related the number of faces, edges and vertices of any polyhedron in one formula.

a Complete:

Solid	Number of Faces (F)	Number of Vertices (V)	Number of Edges (E)	$F+V$
Rectangular Prism		8		
Triangular Prism		6		
Hexagonal Prism		12		
Triangular Pyramid		4		
Pentagonal Pyramid		6		

b Determine whether each of the following rules correctly states the relationship between the number of faces, vertices and edges of a polyhedron:

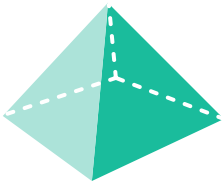
i $F+V-E=2$ ii $F+V+E=2$

iii $E+V-F=2$ iv $F+E-V=2$

20 Select the right name for each solid from the list below:

- Square pyramid
- Rectangular pyramid
- Triangular prism
- Rectangular prism
- Octagonal pyramid
- Octagonal prism

a



b



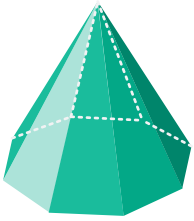
c



d



e



f



21 How many planes of symmetry do the following solids have?

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a Cube

b Sphere

c Cone

d Square prism

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